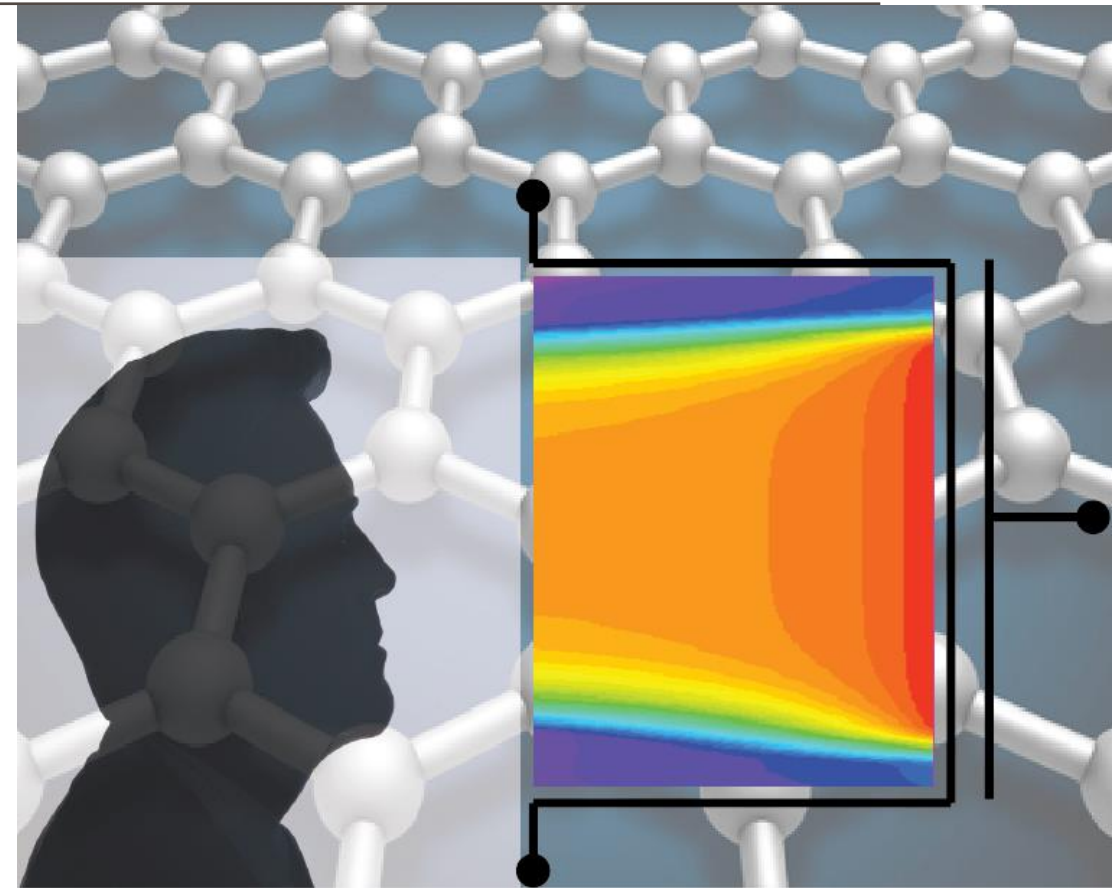


ADVANCED SOLID-STATE DEVICES

Week-4, Lecture-2

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30th January, 2025



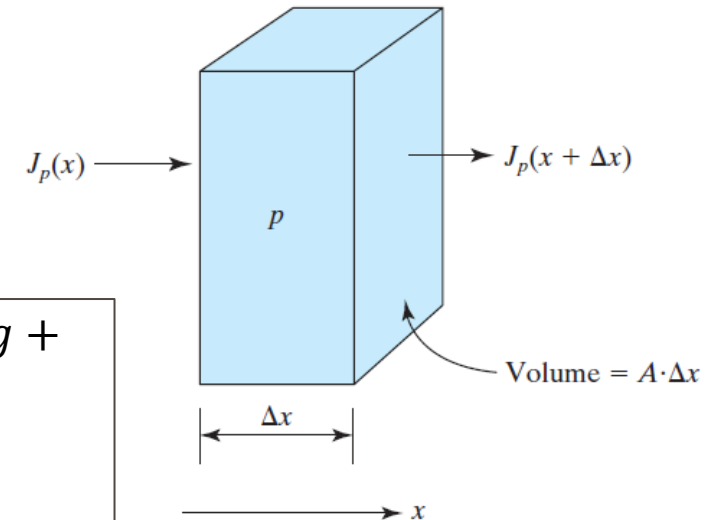
Non-equilibrium Excess Carriers



Continuity
Equations

Continuity Equation (1)

- Consider a box with cross sectional area A and infinitesimal small length Δx . Assume that the incoming hole current density is $J_p(x)$ and the outgoing hole current density is $J_p(x + \Delta x)$. Assume that the concentration of hole is p .



- We can write for the holes: *#increase = # entering – #leaving + #generating – #recombining*
- We can also write in terms of rate:
rate of change in number of holes
= hole entering rate – hole leaving rate + generation rate
– recombination rate

➤ Let us write a differential equation to represent above behavior.

$$A \cdot \Delta x \frac{\partial p}{\partial t} = \frac{J_p}{q} A - \frac{J_p(x+\Delta x)}{q} A + G A \cdot \Delta x - R A \cdot \Delta x$$

$$\Delta x \frac{\partial p}{\partial t} = \frac{J_p}{q} - \frac{J_p(x + \Delta x)}{q} + G \Delta x - \frac{p \Delta x}{\tau_p}$$

Continuity Equation (2)

$$\Delta x \frac{\partial p}{\partial t} = \frac{J_p}{q} - \frac{J_p(x + \Delta x)}{q} + G\Delta x - \frac{p\Delta x}{\tau_p}$$

$$\frac{\partial p}{\partial t} = \frac{J_p - J_p(x + \Delta x)}{q\Delta x} + G - \frac{p}{\tau_p}$$

$$J_p(x + \Delta x) = J_p(x) + \frac{\partial J_p(x)}{\partial x} \Delta x$$

$$\Rightarrow \frac{J_p - J_p(x + \Delta x)}{\Delta x} = -\frac{\partial J_p(x)}{\partial x}$$

$$\frac{\partial p}{\partial t} = -\frac{1}{q} \frac{\partial J_p(x)}{\partial x} + G - \frac{p}{\tau_p}$$

Similarly for electrons (charge is $-q$):

$$\frac{\partial n}{\partial t} = \frac{1}{q} \frac{\partial J_n(x)}{\partial x} + G - \frac{n}{\tau_n}$$

- To solve these equations, we need to write current densities in terms of carrier concentration.
- Writing these equations in term of carrier concentrations gives *carrier transport equations*.

Non-equilibrium Excess Carriers



Carrier Transport
Equations

Carrier Transport Equation

$$\frac{\partial p}{\partial t} = -\frac{1}{q} \frac{\partial J_p(x)}{\partial x} + G - \frac{p}{\tau_p}$$

$$\frac{\partial n}{\partial t} = \frac{1}{q} \frac{\partial J_n(x)}{\partial x} + G - \frac{n}{\tau_n}$$

- Current density is:

$$J = J_p + J_n = qp\mu_p E_x - qD_p \frac{dp}{dx} + qn\mu_n E_x + qD_n \frac{dn}{dx}$$

Putting $J_p = qp\mu_p E_x - qD_p \frac{dp}{dx}$ in $\frac{\partial p}{\partial t} = -\frac{1}{q} \frac{\partial J_p(x)}{\partial x} + G - \frac{p}{\tau_p}$

$$\frac{\partial p}{\partial t} = -\mu_p p \frac{\partial E_x}{\partial x} - \mu_p E_x \frac{\partial p}{\partial x} + D_p \frac{\partial^2 p}{\partial x^2} + G - \frac{p}{\tau_p}$$

Similarly for electrons:

$$\frac{\partial n}{\partial t} = \mu_n n \frac{\partial E_x}{\partial x} + \mu_n E_x \frac{\partial n}{\partial x} + D_n \frac{\partial^2 n}{\partial x^2} + G - \frac{n}{\tau_n}$$

Carrier Transport Equation: Excess Carriers (2)

$$\frac{\partial p}{\partial t} = -\mu_p p \frac{\partial E_x}{\partial x} - \mu_p E_x \frac{\partial p}{\partial x} + D_p \frac{\partial^2 p}{\partial x^2} + G - \frac{p}{\tau_p}$$

Assume p_o is not changing with space, we can write carrier transport equation in terms of excess carriers also.

$$\frac{\partial \delta p}{\partial t} = -\mu_p p \frac{\partial E_x}{\partial x} - \mu_p E_x \frac{\partial \delta p}{\partial x} + D_p \frac{\partial^2 \delta p}{\partial x^2} + G - \frac{p}{\tau_p}$$

Similarly for electrons:

$$\frac{\partial \delta n}{\partial t} = \mu_n n \frac{\partial E_x}{\partial x} + \mu_n E_x \frac{\partial \delta n}{\partial x} + D_n \frac{\partial^2 \delta n}{\partial x^2} + G - \frac{n}{\tau_n}$$

- Note that:

$$n = n_o + \delta n$$

$$p = p_o + \delta p$$

- Carrier transport equation has three unknowns: p , n , and E
- We can take help of Poisson's equation for computation.

Non-equilibrium Excess Carriers



Poisson's Equation

Poisson's Equation

- Poisson's equation is useful for finding the electric potential distribution (and electric field gradient) using carrier concentration.

Poisson's Equation:

$$\nabla^2 \phi = -\frac{\rho}{\epsilon_s}$$

- ϕ is the potential
- ρ is the charge density
- ϵ_s is the semiconductor permittivity

We know that electric field:

$$E = -\nabla \phi$$

So, Poisson's equation is equivalent to: $\nabla \cdot E = \frac{\rho}{\epsilon_s}$

For 1-D we can write:

$$\frac{dE}{dx} = -\frac{d^2 \phi}{dx^2} = \frac{\rho}{\epsilon_s}$$

Non-equilibrium Excess Carriers



Ambipolar
Transport

Ambipolar Transport (1)

Problem: Assume that a pulse of excess electron-hole pairs are created in a semiconductor. Assume that an electric field is also applied along the X-direction. How will the excess carriers move inside the semiconductor.

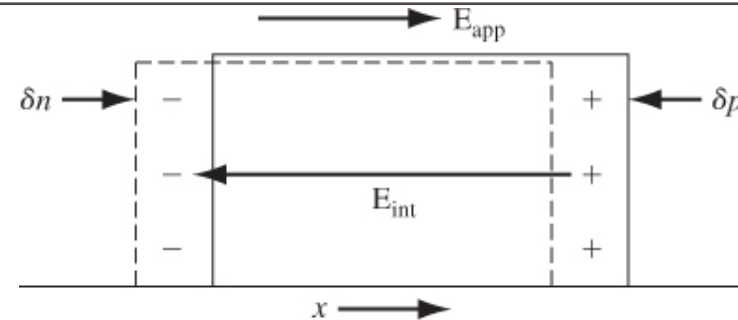


Figure 6.5 | The creation of an internal electric field as excess electrons and holes *tend* to separate.

- Excess electrons and holes will move in opposite directions due to applied field
- Because of separation of charge, an internal electric field is induced
- Internal electric field will create an attraction between excess electrons and holes
- Excess electrons and holes will drift/diffuse together with a *single effective mobility* and *single effective diffusion co-efficient*
- This phenomenon is called ambipolar transport

Ambipolar Transport: Assuming Quasi-neutrality

We assume $\delta p = \delta n$ [Quasi-neutrality condition?] because a small change in carrier concentration is sufficient to create the internal electric field normally encountered in devices.

$$\frac{\partial \delta p}{\partial t} = -\mu_p p \frac{\partial E_x}{\partial x} - \mu_p E_x \frac{\partial \delta p}{\partial x} + D_p \frac{\partial^2 \delta p}{\partial x^2} + G - R$$

$$\frac{\partial \delta n}{\partial t} = \mu_n n \frac{\partial E_x}{\partial x} + \mu_n E_x \frac{\partial \delta n}{\partial x} + D_n \frac{\partial^2 \delta n}{\partial x^2} + G - R$$

Eliminating term of electric field gradient:

$$\frac{\partial \delta n}{\partial t} = -\mu_p p \frac{\partial E_x}{\partial x} - \mu_p E_x \frac{\partial \delta n}{\partial x} + D_p \frac{\partial^2 \delta n}{\partial x^2} + G - R$$

Multiply by $\mu_n n$

$$\frac{\partial \delta n}{\partial t} = \mu_n n \frac{\partial E_x}{\partial x} + \mu_n E_x \frac{\partial \delta n}{\partial x} + D_n \frac{\partial^2 \delta n}{\partial x^2} + G - R$$

Multiply by $\mu_p p$ And add two equations

$$(\mu_n n + \mu_p p) \frac{\partial \delta n}{\partial t} = (\mu_n \mu_p p - \mu_n \mu_p n) E_x \frac{\partial \delta n}{\partial x} + (D_p \mu_n n + D_n \mu_p p) \frac{\partial^2 \delta n}{\partial x^2} + (\mu_n n + \mu_p p) [G - R]$$

Ambipolar Transport Equation

$$(\mu_n n + \mu_p p) \frac{\partial \delta n}{\partial t} = (\mu_n \mu_p p - \mu_n \mu_p n) E_x \frac{\partial \delta n}{\partial x} + (D_p \mu_n n + D_n \mu_p p) \frac{\partial^2 \delta n}{\partial x^2} + (\mu_n n + \mu_p p) [G - R]$$

Divide by $(\mu_n n + \mu_p p)$

$$\frac{\partial \delta n}{\partial t} = \frac{(\mu_n \mu_p p - \mu_n \mu_p n)}{(\mu_n n + \mu_p p)} E_x \frac{\partial \delta n}{\partial x} + \frac{(D_p \mu_n n + D_n \mu_p p)}{(\mu_n n + \mu_p p)} \frac{\partial^2 \delta n}{\partial x^2} + G - R$$

Comparing with transport equation for electrons/holes:

$$\frac{\partial \delta n}{\partial t} = \mu_n n \frac{\partial E_x}{\partial x} + \mu_n E_x \frac{\partial \delta n}{\partial x} + D_n \frac{\partial^2 \delta n}{\partial x^2} + G - R$$

Ambipolar Transport Equation:

$$\frac{\partial \delta n}{\partial t} = \mu' E_x \frac{\partial \delta n}{\partial x} + D' \frac{\partial^2 \delta n}{\partial x^2} + G - R$$

Ambipolar Mobility:

$$\mu' = \frac{(\mu_n \mu_p p - \mu_n \mu_p n)}{(\mu_n n + \mu_p p)}$$

Ambipolar Diffusion Coefficient:

$$D' = \frac{(D_p \mu_n n + D_n \mu_p p)}{(\mu_n n + \mu_p p)}$$

Ambipolar Transport: Extrinsic Doping and Low Level Injection (p-type) (1)

Ambipolar Mobility:

$$\mu' = \frac{\mu_n \mu_p (p - n)}{(\mu_n n + \mu_p p)}$$

Ambipolar Diffusion Coefficient:

$$D' = \frac{(D_p \mu_n n + D_n \mu_p p)}{(\mu_n n + \mu_p p)}$$

Ambipolar Transport Equation:

$$\frac{\partial \delta n}{\partial t} = \mu' E_x \frac{\partial \delta n}{\partial x} + D' \frac{\partial^2 \delta n}{\partial x^2} + G - R$$

Problem:

Assume that a semiconductor material is extrinsic p-type and low-level injection occurs. Simplify the expressions for ambipolar mobility and ambipolar diffusion coefficient based on these assumptions.

p-type extrinsic material: $p_o \gg n_o$

Low-level injection: $p_o \gg \delta n$

Ambipolar Mobility:

$$\mu' = \frac{\mu_n \mu_p (p - n)}{(\mu_n n + \mu_p p)} \approx \frac{\mu_n \mu_p p_o}{\mu_p p_o} = \mu_n$$

Ambipolar Diffusion Coefficient:

$$D' = \frac{(D_p \mu_n n + D_n \mu_p p)}{(\mu_n n + \mu_p p)} \approx \frac{D_n \mu_p p_o}{\mu_p p_o} = D_n$$

The mobility and diffusion coefficient are decided by minority carriers.

Ambipolar Transport: Extrinsic Doping and Low Level Injection (p-type) (2)

Ambipolar Transport Equation:

$$\frac{\partial \delta n}{\partial t} = \mu' E_x \frac{\partial \delta n}{\partial x} + D' \frac{\partial^2 \delta n}{\partial x^2} + G - R$$

At equilibrium : $G_{n0} = R_{n0}$

In a p-type material:

- The holes are plenty, and there are less electrons.
- The probability of finding a hole for an electron remains constant over time
- Lifetime of minority carrier electron will remain constant

We can write the term $G - R$ for electrons as:

$$G - R = G_{n0} + g'_n - (R_{n0} + R'_n)$$

So, we can write: $G - R = g'_n - \frac{\delta n}{\tau_{n0}}$ [in terms of minority carrier]

For p-type material we should write ambipolar transport equation for electrons:

$$\frac{\partial \delta n}{\partial t} = \mu' E_x \frac{\partial \delta n}{\partial x} + D' \frac{\partial^2 \delta n}{\partial x^2} + g'_n - \frac{\delta n}{\tau_{n0}}$$

Ambipolar Mobility:

$$\mu' = \mu_n$$

Ambipolar Diffusion Coefficient: $D' = D_n$

Ambipolar Transport: Extrinsic Doping and Low Level Injection (n-type)

For n-type material we should write ambipolar transport equation for holes:

$$\frac{\partial \delta p}{\partial t} = \mu' E_x \frac{\partial \delta p}{\partial x} + D' \frac{\partial^2 \delta p}{\partial x^2} + g'_n - \frac{\delta p}{\tau_{p0}}$$

Ambipolar Mobility: $\mu' = -\mu_p$

Ambipolar Diffusion Coefficient: $D' = D_p$

Ambipolar Transport Equations: Simplifications

For p-type material :

$$\frac{\partial \delta n}{\partial t} = \mu_n E_x \frac{\partial \delta n}{\partial x} + D_n \frac{\partial^2 \delta n}{\partial x^2} + g'_n - \frac{\delta n}{\tau_{n0}}$$

For n-type material :

$$\frac{\partial \delta p}{\partial t} = -\mu_p E_x \frac{\partial \delta p}{\partial x} + D_p \frac{\partial^2 \delta p}{\partial x^2} + g'_p - \frac{\delta p}{\tau_{p0}}$$

Table 6.2 | Common ambipolar transport equation simplifications

Specification	Effect
Steady state	$\frac{\partial(\delta n)}{\partial t} = 0, \quad \frac{\partial(\delta p)}{\partial t} = 0$
Uniform distribution of excess carriers (uniform generation rate)	$D_n \frac{\partial^2(\delta n)}{\partial x^2} = 0, \quad D_p \frac{\partial^2(\delta p)}{\partial x^2} = 0$
Zero electric field	$E \frac{\partial(\delta n)}{\partial x} = 0, \quad E \frac{\partial(\delta p)}{\partial x} = 0$
No excess carrier generation	$g' = 0$
No excess carrier recombination (infinite lifetime)	$\frac{\delta n}{\tau_{n0}} = 0, \quad \frac{\delta p}{\tau_{p0}} = 0$

Ambipolar Transport Equations: An Example (1)

Problem: Consider a p-type semiconductor that is homogeneous and infinite in extent. No external electric field is applied. Assume excess carriers are being generated only at $x = 0$, as shown in the figure.

The excess carrier will start diffusing in $+x$ and $-x$ directions. Calculate the steady-state excess carrier concentration as a function of x .

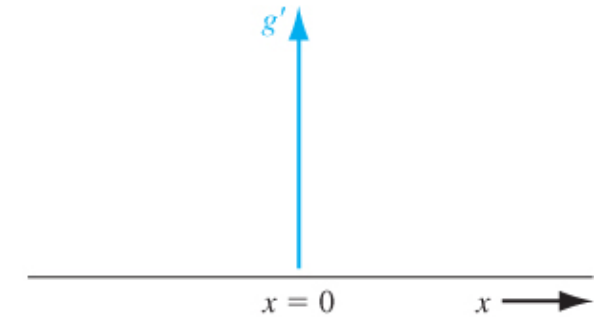


Figure 6.6 | Steady-state generation rate at $x = 0$.

For p-type material :

$$\frac{\partial \delta n}{\partial t} = \mu_n E_x \frac{\partial \delta n}{\partial x} + D_n \frac{\partial^2 \delta n}{\partial x^2} + g'_n - \frac{\delta n}{\tau_{n0}}$$

Simplifies to:

$$D_n \frac{\partial^2 \delta n}{\partial x^2} - \frac{\delta n}{\tau_{n0}} = 0$$

We define diffusion length:

$$L_n = \sqrt{D_n \tau_{n0}}$$

For steady state: $\frac{\partial \delta n}{\partial t} = 0$

No electric field: $E_x = 0$

For $x \neq 0$, $g'_n = 0$

Dividing by D_n :

$$\frac{\partial^2 \delta n}{\partial x^2} - \frac{\delta n}{D_n \tau_{n0}} = 0$$

Equation becomes:

$$\frac{\partial^2 \delta n}{\partial x^2} - \frac{\delta n}{L_n^2} = 0$$

Ambipolar Transport Equations: An Example (2)

The solution of the equation: $\frac{\partial^2 \delta n}{\partial x^2} - \frac{\delta n}{L_n^2} = 0$ is of the form:

$$\delta n = Ae^{-x/L_n} + Be^{x/L_n}$$

- A and B are constants.
- We can apply boundary conditions to determine these constants.

For $x \geq 0$: $\delta n = 0$ for $x = \infty$

$$\Rightarrow B = 0$$

$$\delta n(x) = Ae^{-x/L_n}$$

$$\delta n(x) = \delta n(0)e^{-x/L_n}$$

For $x \leq 0$: $\delta n = 0$ for $x = -\infty$

$$\Rightarrow A = 0$$

$$\delta n(x) = Be^{x/L_n}$$

$$\delta n(x) = \delta n(0)e^{x/L_n}$$

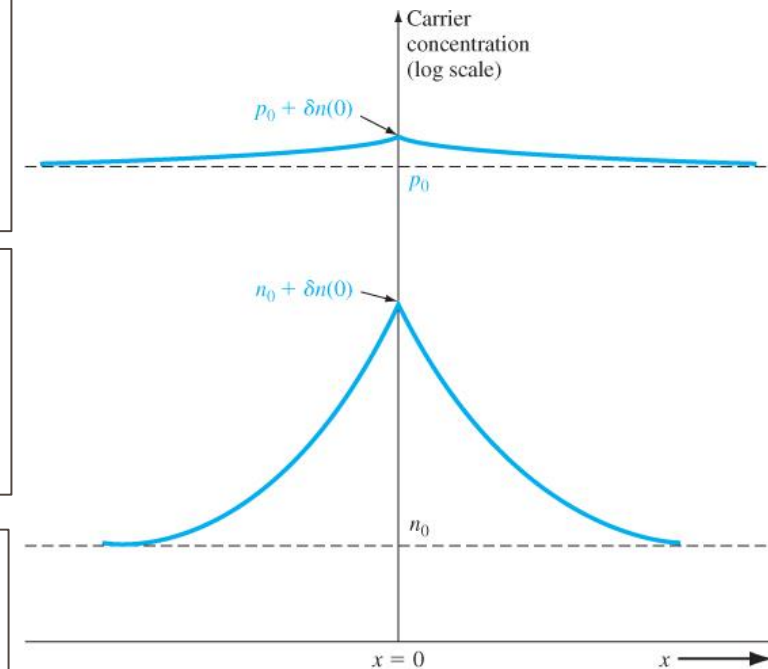


Figure 6.7 | Steady-state electron and hole concentrations for the case when excess electrons and holes are generated at $x = 0$.

Do more problems on carrier transport equation.

PN Junction

PN Junction: Structure

- A pn junction exists in most semiconductor devices
- A pn junction is formed when p-type and n-type regions are adjacent *on the same crystal*
- For example, if we dope two regions with acceptors and donors with a sharp boundary between two.

- A concentration gradient exists in a pn junction
- Leads to diffusion of carriers

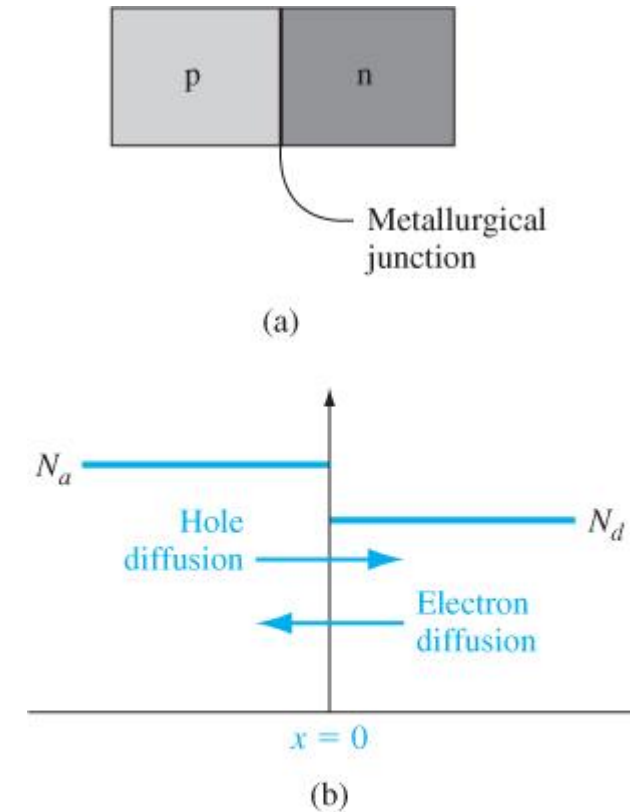


Figure 7.1 | (a) Simplified geometry of a pn junction; (b) doping profile of an ideal uniformly doped pn junction.

PN Junction: At Equilibrium

PN Junction: At Equilibrium

- Holes diffuse from the p-side leaving behind negative charged ions
- Electrons diffuse from the n-side leaving behind positive charged ions
- Electric field is established
- The magnitude of electric field is such that “diffusion force” is balanced

Space charge or Depletion Region

- Around the junction, only immobile ions exist, no free carriers (similar to insulator)
- Separates two p-type and n-type regions

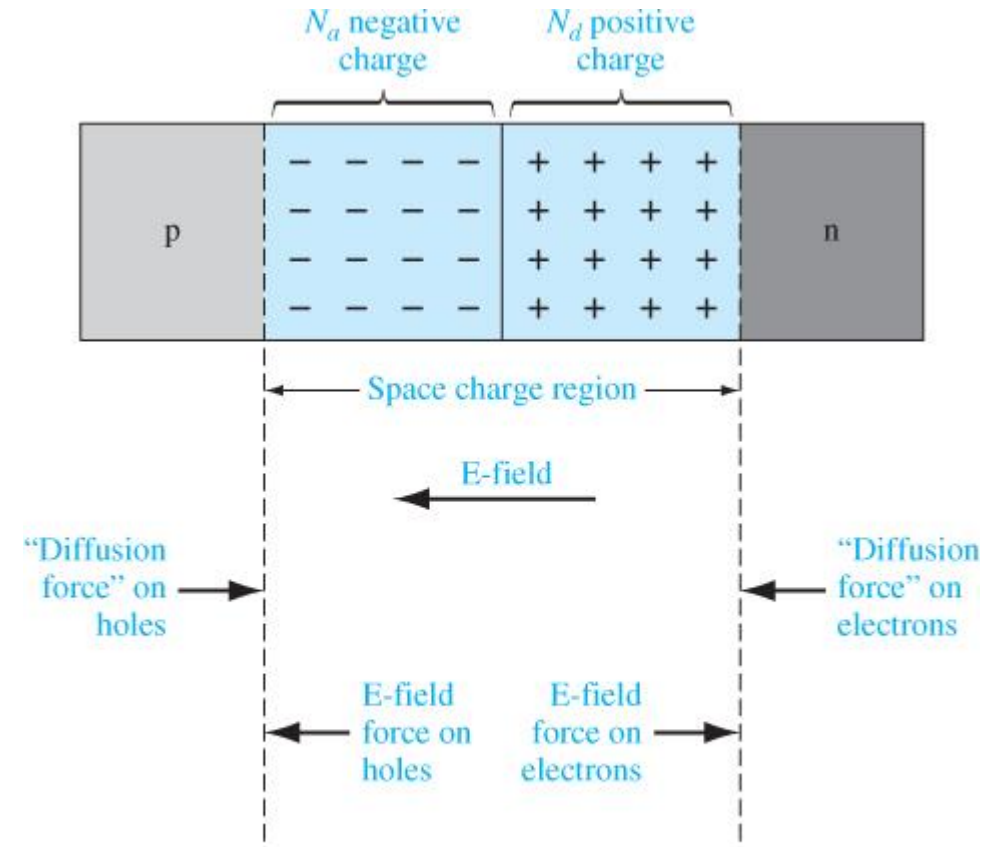


Figure 7.2 | The space charge region, the electric field, and the forces acting on the charged carriers.

PN Junction: Energy Band Diagram

Problem: Draw the energy band diagram at equilibrium, for a pn junction with step-type doping profile

Following points can be used to draw the energy band diagram

- At equilibrium, the Fermi energy level is flat
- At p-side, the Fermi level is close to the valence band
- At n-side, the Fermi level is close to the conduction band

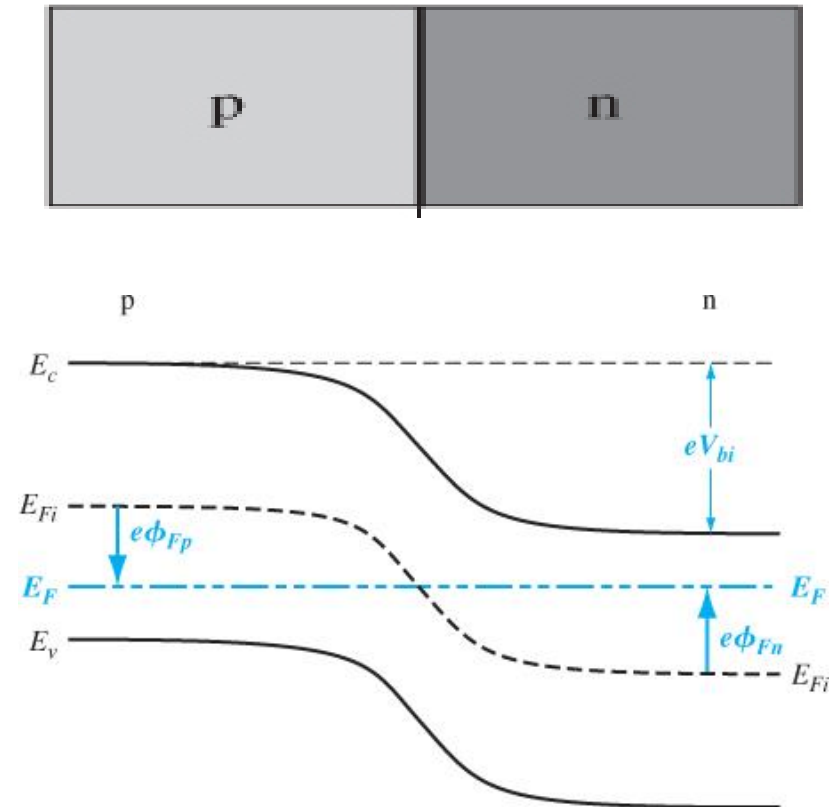


Figure 7.3 | Energy-band diagram of a pn junction in thermal equilibrium.

PN Junction: Built-in Potential Barrier (1)

- Due to band-bending, the electron in the conduction band on the n-side see a barrier in moving to the p-side.
- This barrier is known as built-in potential barrier.

Problem: Compute the magnitude of the potential barrier using the energy band diagram (in terms of Volts)

- We need to find the energy by which the conduction band edge E_C has moved.
- E_C and E_{Fi} are equidistant along the junction
- We can use E_{Fi} for computation

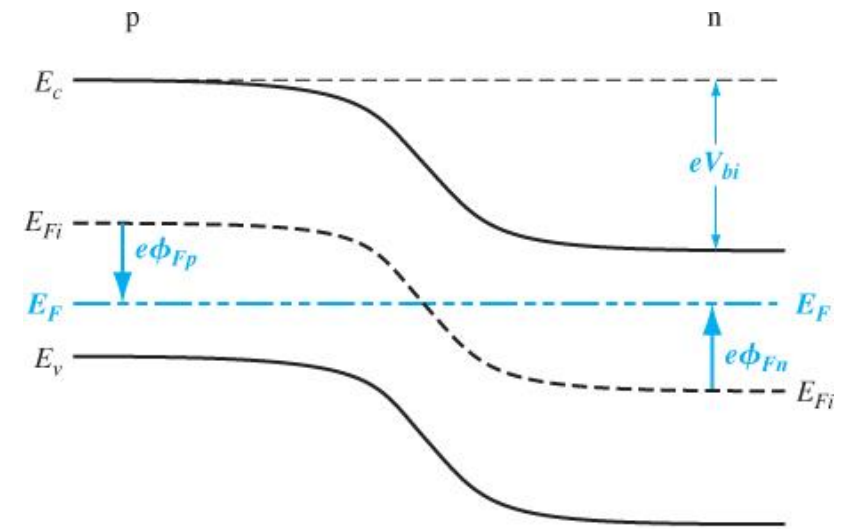


Figure 7.3 | Energy-band diagram of a pn junction in thermal equilibrium.

PN Junction: Built-in Potential Barrier (2)

Considering hole concentration on the p-side:

$$p_0 = N_a = n_i \exp \left[\frac{E_{Fi} - E_F}{kT} \right]$$

$$p_0 = N_a = n_i \exp \left[\frac{q\phi_{Fp}}{kT} \right]$$

$$\phi_{Fp} = \frac{kT}{q} \ln \left(\frac{N_a}{n_i} \right)$$

Similarly, considering electron concentration on the n-side

$$\phi_{Fn} = -\frac{kT}{q} \ln \left(\frac{N_d}{n_i} \right)$$

Built-in potential is:

$$V_{bi} = |\phi_{Fp}| + |\phi_{Fn}| = \frac{kT}{q} \ln \left(\frac{N_a N_d}{n_i^2} \right)$$

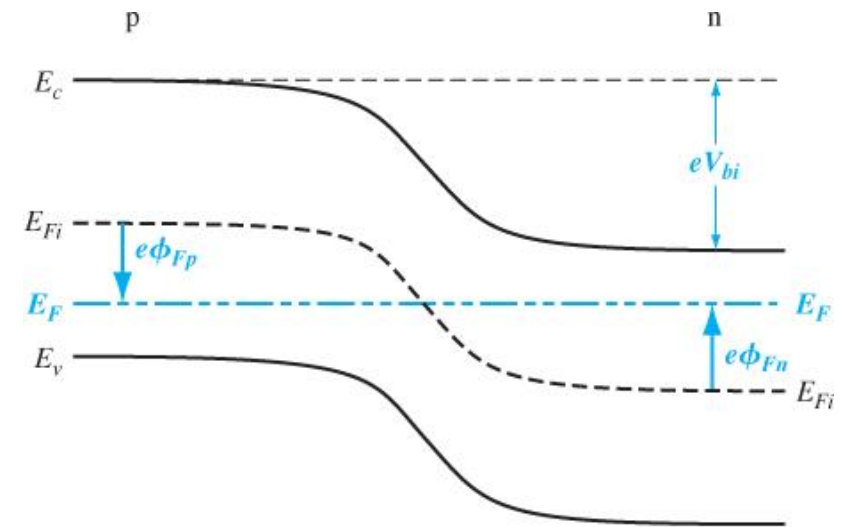


Figure 7.3 | Energy-band diagram of a pn junction in thermal equilibrium.

- We refer to $\frac{kT}{q}$ as thermal voltage and denote as V_t

$$V_{bi} = V_t \ln \left(\frac{N_a N_d}{n_i^2} \right)$$

References

- D. A. Neaman, *"Semiconductor Physics and Devices"*